

PLANTNEST PROJECT SYNOPSIS

1. TITLE OF THE PROJECT

PlantNest: An Online Plant Store and Community Platform for Plant Lovers

2. INTRODUCTION

PlantNest is a web-based platform developed to help users discover, purchase, and learn to care for indoor plants.

3. PROBLEM STATEMENT

Many people want to keep plants at home, but they often face difficulties in selecting suitable plants, understanding their care requirements, and finding reliable sources for purchase.

4. OBJECTIVES

The main objectives of PlantNest are:

- To provide an online platform for browsing and purchasing a variety of plants.
- To allow users to save favorite plants in a wishlist for future purchase.
- To help users place and track plant orders easily.
- To guide beginners with simple maintenance tips and plant care suggestions.
- To create a community page where plant lovers can share their plant journey, experiences, and joy.
- To deliver a responsive and visually appealing user interface for both desktop and mobile users.

5. SCOPE OF THE PROJECT

The scope of PlantNest includes product browsing, filtering, sorting, wishlist management, checkout, and order tracking.

6. EXISTING SYSTEM

In the existing system, many online stores only provide product listings and payment options. Users struggle to find detailed care information and a supportive plant community.

7. PROPOSED SYSTEM

The proposed system, PlantNest, integrates plant shopping, care guidance, and community interaction into a single, user-friendly platform.

8. MODULES OF THE PROJECT

The major modules of PlantNest are:

8.1 Plant Catalog Module

Displays a variety of plants with details such as name, category, light requirement, size, price, and description.

8.2 Search, Filter, and Sort Module

Allows users to search plants by name or category and filter them by size, light requirement, and price range.

8.3 Wishlist Module

Enables users to save favorite plants for future purchase and manage their saved collection.

8.4 Checkout and Order Module

Allows users to enter delivery details, select payment methods, place orders, and view order history.

8.5 Plant Care Guide Module

Provides beginner-friendly guidance on watering, lighting, common plant problems, and simple maintenance tips.

8.6 Community Module

Offers a dedicated space where plant lovers can post updates, share their plant journey, and interact with others.

8.7 Authentication Module

Provides basic sign-up and sign-in functionality for personalized usage.

9. TECHNOLOGIES USED

Frontend:

- React.js
- React Router
- CSS
- React Icons

Backend:

- Node.js
- Express.js

Database:

- MongoDB
- Mongoose

Other Tools:

- Axios
- Vite

10. SYSTEM WORKING

The user opens the PlantNest application and browses the available plants. Using search, filters, and sorting options to find the desired plant.

11. ADVANTAGES OF THE PROJECT

- User friendly and visually engaging interface.
- Combines shopping, learning, and community interaction in one platform.
- Beginner-friendly guidance for plant maintenance.
- Organized browsing with filters and sorting features.
- Encourages engagement and emotional connection through the community space.

12. LIMITATIONS

- The current authentication system is basic and can be improved.
- The community feature currently uses lightweight local storage behavior and can be extended with shared database.
- Online payment gateway integration is not fully implemented.
- Admin management features are not included in the current version.

13. FUTURE ENHANCEMENTS

- Add secure user authentication and authorization.
- Implement real-time community chat and commenting.
- Add admin dashboard for managing plants, users, and orders.
- Integrate online payment gateways.
- Add image upload support for community posts.
- Add recommendation features based on plant care level and user interest.
- Enable shared community posts using backend database storage.

14. CONCLUSION

PlantNest is a practical and modern web application that goes beyond a normal plant store by creating